

Geometry-Aware Deep Congruence Networks for Manifold Learning in Cross Subject Motor Imagery

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- ▶ Motor-Imagery EEG (MI-EEG) enables BCI control for motor-impaired users.
- ▶ Applications include neuro-rehabilitation, assistive robotics, and hands-free interaction.
- ▶ EEG signals are low-SNR, non-stationary, and exhibit high inter-subject variability.
- ▶ Recent work suggests covariance-based representations capture EEG structure well and are effective for Motor Imagery decoding.

Problem Statement

- ▶ Given an EEG trial of a subject represented by the covariance matrix, our objective is to train a model, that can perform zero shot action classification for an unseen test subject.
- ▶ We use the **Zero-shot LOSO** evaluation metric: train on all subjects except one and test on the held-out subject without any target labels.
- ▶ Subject variability leads to covariance shifts and orientation & dispersion mismatches, causing action-specific data to overlap or scatter across subjects.
- ▶ Deep models rely on target-subject fine-tuning → fail in true zero-shot settings.
- ▶ Riemannian Alignment reduces mean-shift but cannot correct deeper orientation/dispersion distortions.



Overview

- ▶ I propose a unified geometry-aware framework that optimizes distortions by reducing subject-specific scatter and enhancing action-specific structure, making MI-EEG covariances easier to classify and generalize among unseen subjects.

Learnable Riemannian Pre-Aligners

- ▶ **DCR** — optimizes rotation to improve action discriminability.
- ▶ **RiFU** — U-Net style SPD encoder–decoder congruence transforms to increase the subject scatter to reduce discriminability and increase action discriminability.

Congruence-Based Deep Classifiers

- ▶ **RiFUNet** — U-Net style SPD encoder–decoder with a tangent-space classifier constrained using action, subject-specific distortion and reconstruction loss.
- ▶ **SPD-DCNet** — flexible stack of congruence transforms on SPD manifolds similar to RiFUNet Classifier and without reconstruction constraints.

Key Advantages

- ▶ Learns hierarchical SPD transformations that corrects mean-shift, orientation, and dispersion mismatches jointly
- ▶ Achieves subject-invariant, calibration free zero-shot decoding.



We use the BCI-IV 2a motor-imagery dataset, which contains EEG from **9 subjects** recorded with **22 EEG** and 3 EOG (Electro-oculogram) channels.

- ▶ Two sessions per subject, **288 trials** each
- ▶ Four actions: left hand, right hand, both feet, tongue
- ▶ Each trial: 7–8 s (2 s cue + 4 s MI)
- ▶ Sampling rate: 250 Hz
- ▶ Total per session: ~ 2000 – 2400 s of EEG

Let $X_{i,a,e} \in \mathbb{R}^{C \times T_e}$ denote the EEG epoch for subject i , action a , and trial e , where $C = 22$ channels and T_e is the epoch duration (typically 3–4 s after cue onset). Each trial has a label $y_{i,e} \in \{0, 1, 2, 3\}$ corresponding to the four MI classes.

$$C_{i,a,e} = \frac{1}{T_e - 1} X_{i,a,e} X_{i,a,e}^\top$$

This yields an SPD covariance $C_{i,a,e} \in \mathcal{S}_{++}^C$, which serves as the core representation for all preprocessing and classification stages.



SPD Manifold and Geometry

- ▶ EEG covariance matrices lie on the manifold of **Symmetric Positive Definite (SPD)** matrices. This space is *curved*, not Euclidean.
- ▶ Operations like averaging, distance & interpolation must respect manifold geometry. Euclidean methods distort covariance structure $\rightarrow$ poor alignment.

AIRM Geodesic Distance

The Affine-Invariant Riemannian Metric (AIRM) defines the natural distance on the SPD manifold:

$$d_{\text{AIRM}}(C_1, C_2) = \|\log(C_1^{-1/2} C_2 C_1^{-1/2})\|_F.$$

Key intuition:

- ▶ We **whiten** C_1 by mapping it to the Identity: $C_1^{-1/2} C_1 C_1^{-1/2} = I$.
- ▶ This transforms C_2 into the coordinate system centered at C_1 .
- ▶ At Identity, the SPD manifold becomes **flat** (tangent space).
- ▶ The **logarithm** maps C_2 into this tangent space.
- ▶ The Frobenius norm in tangent space is **isometric** to geodesic distance on the SPD manifold.



Log and Exp Maps

To process SPD matrices in Euclidean space, we use:

$$\log_C(C_2) = C^{1/2} \log(C^{-1/2} C_2 C^{-1/2}) C^{1/2},$$

$$\exp_C(V) = C^{1/2} \exp(C^{-1/2} V C^{-1/2}) C^{1/2}.$$

Interpretation:

- ▶ $\log_C$ projects matrices to the tangent-space at C .
- ▶ $\exp_C$ maps tangent vectors back to manifold.
- ▶ Enables linear learning in tangent space while respecting SPD geometry.

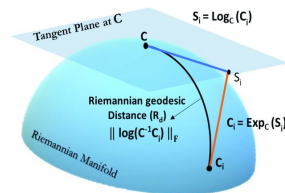


Figure 1: Log/Exp map intuition



Congruence Transforms on SPD Matrices

A fundamental operation in our models is the *congruence transform*:

$$C' = WCW^\top, \quad W \in \mathbb{R}^{d \times C} \text{ full-rank.}$$

Lemma (SPD Preservation)

For any $C \in S_{++}^C$ and full-rank W , the matrix $C' = WCW^\top$ remains SPD:

$$x^\top C' x = (W^\top x)^\top C (W^\top x) > 0 \quad \forall x \neq 0.$$

Lemma (Affine Invariance)

Congruence transforms preserve the AIRM distance:

$$d_{\text{AIRM}}(C_1, C_2) = d_{\text{AIRM}}(WC_1W^\top, WC_2W^\top).$$

These results show that congruence transforms are **isometries** on the SPD manifold, preserving positive-definiteness and AIRM geometry.



Riemannian Alignment (RA)

- ▶ Given a reference covariance C_{ref} , whitening seeks a matrix P satisfying

$$PC_{\text{ref}}P^{\top} = I.$$

- ▶ Under a congruence transform $PCP^{\top}$, the unique SPD solution is $P = C_{\text{ref}}^{-1/2}$, mapping C_{ref} to identity. This aligns all covariances around C_{ref} as

$$C'_{i,a,e} = C_{\text{ref}}^{-1/2} C_{i,a,e} C_{\text{ref}}^{-1/2}.$$

- ▶ RA removes subject-specific mean shifts and establishes a geometric reference,
- ▶ **Limitation:** It does not correct orientation differences or inter-subject dispersion, motivating learned alignment methods.



Preprocessing Methods



Common Spatial Patterns (CSP)

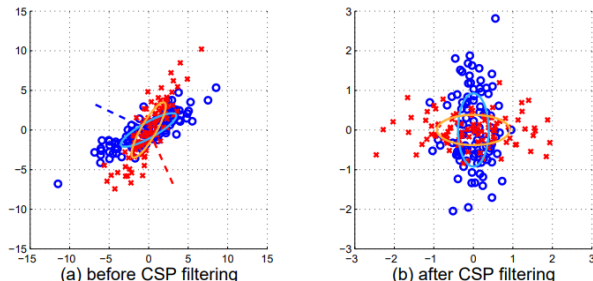


Figure 2: Common Spatial Patterns (CSP): Before filtering (left) the two Gaussian classes have overlapping correlated structure.

- ▶ CSP rotates the projection axes so that each class becomes uncorrelated and attains maximum variance along its own principal direction.
- ▶ **Limitation:** Although discriminative, CSP is highly sensitive to subject-specific covariance structure and degrades under cross-subject shifts.



Algorithm 1 Common Spatial Patterns (CSP)

- 1: **Input:** EEG trials X_{ki} for classes $k \in \{0, 1, 2, 3\}$.
- 2: Compute trial covariance: $C_{ki} = \frac{1}{N_k} X_{ki} X_{ki}^\top$.
- 3: Compute class-average covariance: $C_k = \frac{1}{N_k} \sum_i C_{ki}$.
- 4: One-vs-rest covariance: $C_{-k} = \frac{1}{\sum_{j \neq k} N_j} \sum_{j \neq k} \sum_i C_{ji}$; $C_t = C_k + C_{-k}$.
- 5: Maximize Rayleigh quotient by solving the generalized eigenproblem:

$$J(\omega) = \frac{\omega^\top C_k \omega}{\omega^\top C_{-k} \omega}; \quad C_k \omega = \lambda C_t \omega$$

- 6: Whitening transform on both sides: $S_k = C_t^{-1/2} C_k C_t^{-1/2}$; $S_k v = \lambda v$.
- 7: Construct CSP filters with eigenvectors corresponding to top-m and bottom-m of sorted eigenvalues: $W_{\text{CSP}} = C_t^{-1/2} [v_1, \dots, v_m, v_{C-m+1}, \dots, v_C]$.
- 8: Project EEG: $Z_{ki} = W_{\text{CSP}}^\top X_{ki}$.
- 9: Log-variance features:

$$f(Z_{ki}, c) = \log \frac{\text{var}(Z_{ki}(c, :))}{\sum_{c'} \text{var}(Z_{ki}(c', :))}.$$

- 10: **Output:** CSP feature vector $f(Z_{ki}, c)$.
-



Linear Discriminant Analysis (LDA)

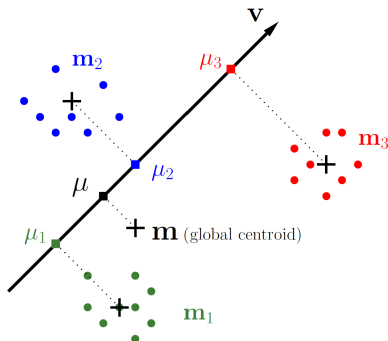


Figure 3: LDA visualization showing class centroids m_1, m_2, m_3 , their projected means μ_1, μ_2, μ_3 , and the discriminant direction v along which between-class variance is maximized relative to within-class variance.

- **Limitation:** LDA improves discriminability but remains linear and cannot capture nonlinear or geometric subject variability.



Algorithm 2 Linear Discriminant Analysis (LDA)

- 1: **Input:** Tangent-space vectors $z_i \in \mathbb{R}^d$ with class labels y_i .
- 2: Compute class means and global mean (inline): $\mu_k = \frac{1}{|\mathcal{C}_k|} \sum_{i \in \mathcal{C}_k} z_i$, $\mu = \frac{1}{N} \sum_i z_i$.
- 3: Compute within-class scatter and between-class scatter:

$$S_W = \sum_k \sum_{i \in \mathcal{C}_k} (z_i - \mu_k)(z_i - \mu_k)^\top, \quad S_B = \sum_k |\mathcal{C}_k| (\mu_k - \mu)(\mu_k - \mu)^\top.$$

- 4: Maximize Fisher ratio (objective) which yields the generalized eigenproblem.

$$J(v) = \frac{v^\top S_B v}{v^\top S_W v}; \quad S_B v = \lambda S_W v$$

- 5: Select top $m = K - 1$ eigenvectors; form projection matrix $V = [v_1, \dots, v_m]$.
 - 6: Project samples into LDA space: $z_i^{\text{LDA}} = V^\top z_i$.
 - 7: **Output:** Low-dimensional LDA features z_i^{LDA} and projection matrix V .
-



Riemannian Procrustes Analysis (RPA)

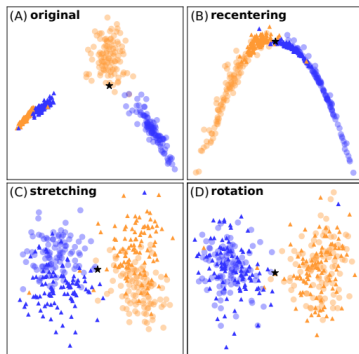


Figure 4: Representation of sequence of operations in RPA re-center $\rightarrow$ scale $\rightarrow$ rotate.

- **Limitation:** RPA reduces global subject-level variation but remains a static, non-learned transform and cannot fully correct class-dependent or fine-grained orientation mismatches



Algorithm 3 Riemannian Procrustes Analysis (RPA) $\rightarrow$ Transfer Learning

- 1: **Input:** Covariances from source $\{C_s\}$ and target $\{C_t\}$
- 2: **Re-center (RA):**
- 3: Compute reference mean C_{ref} ; re-center:

$$C' = C_{\text{ref}}^{-1/2} C C_{\text{ref}}^{-1/2}$$

- 4: **Scale (match dispersions):**

$$\varsigma_s^2 = \frac{1}{|S|} \sum d^2(C'_s, I), \quad \varsigma_t^2 = \frac{1}{|T|} \sum d^2(C'_t, I), \quad \rho = \sqrt{\frac{\varsigma_s^2}{\varsigma_t^2}}$$

- 5: Log-scale target: $Z = \log C'$, $Z' = \rho Z$, $\tilde{C} = \exp(Z')$
- 6: **Rotate (match orientations):**
- 7: Compute class-wise means: $\tilde{C}_{s,k}$, $\tilde{C}_{t,k}$
- 8: Map to tangent: $T_{s,k} = \log \tilde{C}_{s,k}$, $T_{t,k} = \log \tilde{C}_{t,k}$
- 9: Solve:

$$U = \arg \min_{U \in O(d)} \sum_k \|U^\top T_{t,k} U - T_{s,k}\|_F^2$$

- 10: Rotate scaled target: $C^{\text{out}} = U^\top \tilde{C} U$
 - 11: **Output:** Final aligned covariances C^{out}
-



Classifiers



Tangent Space Logistic Regression

- ▶ After Riemannian Alignment, each covariance matrix is first transported to the tangent space at the identity by applying the matrix logarithm

$$T_{i,a,e} = \log(C_{\text{ref}}^{-\frac{1}{2}} C_{i,a,e} C_{\text{ref}}^{-\frac{1}{2}}).$$

- ▶ The symmetric matrix $T_{i,a,e}$ is then vectorized using its upper-triangular entries and directly provided to a multinomial logistic regression classifier:

$$z_{i,a,e} = \text{vec}_{\text{upper}}(T_{i,a,e}),$$

$$p(y = k \mid z_{i,a,e}) = \frac{\exp(w_k^\top z_{i,a,e} + b_k)}{\sum_{j=1}^K \exp(w_j^\top z_{i,a,e} + b_j)}.$$

- ▶ TSLR therefore combines Riemannian geometry (via log and vectorization) with simple linear decision boundaries in Euclidean space.
- ▶ **Limitation:** Its a shallow baseline classifier and cannot capture hierarchical distortions.



Algorithm 4 Tangent Space Logistic Regression (TSLR)

- 1: **Input:** EEG trials $X_{i,a,e}$ with class labels $y_{i,a,e}$.
- 2: Covariance (one-line): $C_{i,a,e} = \frac{1}{T-1} X_{i,a,e} X_{i,a,e}^\top$.
- 3: Subject reference mean (one-line): $C_{\text{ref}}^{(i)} = \text{Riem./Euclid./LogEuc mean.}$
- 4: **Riemannian Alignment (RA):**

$$C'_{i,a,e} = (C_{\text{ref}}^{(i)})^{-1/2} C_{i,a,e} (C_{\text{ref}}^{(i)})^{-1/2}$$

- 5: Global mean (one-line): $C_G = \text{mean}_{\text{Riemann}}(\{C'_{i,a,e}\})$.
- 6: **Tangent projection at C_G :**

$$T_{i,a,e} = C_G^{1/2} \log(C_G^{-1/2} C'_{i,a,e} C_G^{-1/2}) C_G^{1/2}$$

- 7: Vectorize (one-line): $x_{i,a,e} = \text{vec}_\Delta(T_{i,a,e})$.

- 8: **Multinomial Logistic Regression:** $z_k = w_k^\top x + b_k$ and $p_k(x) = \frac{e^{z_k}}{\sum_j e^{z_j}}$.

- 9: Prediction (inline): $\hat{y} = \arg \max_k p_k(x)$.

- 10: **Output:** Predicted label $\hat{y}$ and probabilities $p_k(x)$.
-



Minimum Distance to Mean (MDM)

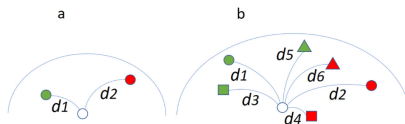


Figure 5: Representation of minimum distance to mean (MDM) classifier on the Riemannian manifold

- **Limitation:** MDM operates directly on the SPD geometry and therefore lacks any learnable transformation to correct subject-specific distortions



Algorithm 5 Minimum Distance to Mean (MDM)

- 1: **Input:** EEG trials $X_{i,a,e}$ for subjects i , actions a , epochs e .
- 2: Covariance estimation (one-line): $C_{i,a,e} = \frac{1}{T-1} X_{i,a,e} X_{i,a,e}^\top$.
- 3: **Compute class means** C_{ref_k} :
- 4: Euclidean mean (inline): $C_{\text{ref}_k} = \frac{1}{N_k} \sum_{i,e:a=k} C_{i,a,e}$.
- 5: Log-Euclidean mean (inline): $C_{\text{ref}_k} = \exp\left(\frac{1}{N_k} \sum_{i,e:a=k} \log C_{i,a,e}\right)$.
- 6: Riemannian (AIRM) mean:

$$C_{\text{ref}_k} = \arg \min_{C \in \mathbb{S}_{++}^C} \sum_{i,e:a=k} \left\| \log\left(C^{-1/2} C_{i,a,e} C^{-1/2}\right) \right\|_F^2$$

- 7: **Distance to class k :**

$$D_k(C) = \left\| \log\left(C^{-1/2} C_{\text{ref}_k} C^{-1/2}\right) \right\|_F$$

- 8: **Predict label (inline):** $\hat{y} = \arg \min_k D_k(C)$.
 - 9: **Output:** Predicted label $\hat{y}$.
-



LDA Classifier

- ▶ When used as a classifier, Linear Discriminant Analysis (LDA) models each class as a Gaussian distribution with a shared covariance matrix.
- ▶ The Bayes-optimal discriminant function for class k is

$$\delta_k(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k,$$

where x is the feature vector, μ_k is the class mean, Σ is the shared within-class covariance, and π_k is the class prior.

- ▶ CSP_LDA applies this discriminant to CSP-filtered features, while TSA_LDA applies it to tangent-space vectors.
- ▶ **Limitation:** Despite their popularity, both remain linear and are sensitive to subject-specific shifts, limiting zero-shot performance.



Preprocessing Module



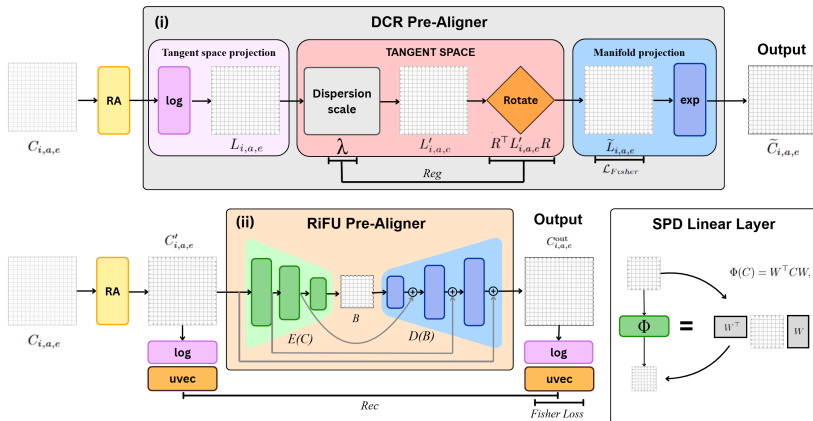


Figure 6: (i) **DCR Pre-Aligner:** Learns dispersion scaling and rotation in the tangent space after RA to produce an aligned SPD. (ii) **RiFU Pre-Aligner:** Uses an RA-initialized Riemannian U-Net to reconstruct an aligned SPD with multi-scale geometric structure.

Discriminative Class Representation (DCR) Pre-Aligner

Problem. Even after subject-wise Riemannian Alignment (RA), RA-whitened EEG covariances $C_{i,a,e} \in \mathbb{S}_{++}^C$ still show:

- ▶ *rotational inconsistencies* across subjects,
- ▶ tilted action-specific directions in the log domain,
- ▶ inflated within-class scatter,
- ▶ degraded zero-shot discrimination.

Idea. DCR learns two geometric corrections in the log-Euclidean domain [?]:

- (i) dispersion scaling λ , (ii) global rotation $R \in SO(C)$,

to realign class structure and enforce consistent action geometry.

Effect. DCR produces a well-centered, well-separated tangent-space representation that better supports zero-shot cross-subject decoding.



Discriminative Class Representation (DCR) Pre-Aligner

Log-Domain Transformation

- ▶ Log Covariance: $L_{i,a,e} = \log(C_{i,a,e})$
- ▶ Dispersion Scaling: $L'_{i,a,e} = \lambda L_{i,a,e}$, where $\lambda = \text{softplus}(\lambda_{\text{raw}}) + 10^{-6}$
- ▶ Apply Global Rotation: $\tilde{L}_{i,a,e} = R^\top L'_{i,a,e} R$, where $R \in SO(C)$, $R^\top R = I$

Class Means

- ▶ Action-specific mean: $M_k = \frac{1}{|\mathcal{I}_k|} \sum_{(i,a,e) \in \mathcal{I}_k} L'_{i,a,e}$
- ▶ Global mean: $M = \frac{1}{N} \sum_k |\mathcal{I}_k| M_k$

Fisher statistics

- ▶ Between-Class Variance (i) and Within-Class Variance (ii):

$$(i) B(R) = \sum_k |\mathcal{I}_k| \|R^\top (M_k - M) R\|_F^2 \quad (ii) W(R) = \sum_{i,a,e} \|R^\top (L'_{i,a,e} - M_{y_{i,a,e}}) R\|_F^2$$

- ▶ Center (Correlation) Penalty

$$C(R) = \left\| \text{offdiag} \left(\frac{1}{N} \sum_{i,a,e} \tilde{L}_{i,a,e} \right) \right\|_F^2$$



Discriminative Class Representation (DCR) Pre-Aligner

► Fisher Discriminative Loss

$$\mathcal{L}_{\text{Fisher}} = \gamma \frac{W(R)}{B(R) + \varepsilon} + \gamma_c C(R)$$

► Regularization

$$Reg = \alpha(\lambda - 1)^2 + \beta_t \frac{\|R - I\|_F^2}{C}, \quad \beta_t = \beta \frac{1 + \cos(\pi t / T)}{2}$$

► Total Loss Function (Objective)

$$\boxed{\mathcal{L}_{\text{DCR}} = \mathcal{L}_{\text{Fisher}} + Reg}$$



Algorithm 6 Discriminative Class Representation (DCR) Pre-Aligner

- 1: **Input:** RA-whitened covariances $C'_{i,a,e}$, labels $y_{i,a,e}$.
 - 2: Log-map each covariance (inline): $L_{i,a,e} = \log C'_{i,a,e}$.
 - 3: Compute action means (inline): $M_k = \frac{1}{|\mathcal{I}_k|} \sum L_{i,a,e}$ for $y = k$.
 - 4: Global mean (inline): $M = \frac{1}{N} \sum_k |\mathcal{I}_k| M_k$.
 - 5: Initialize skew generator $A = 0$; rotation $R = \exp(A - A^\top)$; dispersion scale λ .
 - 6: **Optimization Loop:**
 - 7: Update rotation (inline): $R = \exp(A - A^\top)$.
 - 8: Dispersion scaling (inline): $L'_{i,a,e} = \lambda L_{i,a,e}$.
 - 9: Rotate logs (inline): $\tilde{L}_{i,a,e} = R^\top L'_{i,a,e} R$.
 - 10: Compute Fisher terms $B(R)$, $W(R)$, center penalty $C(R)$.
 - 11: Loss (inline): $\mathcal{L}_{\text{DCR}} = \gamma \frac{W(R)}{B(R) + \varepsilon} + \gamma_c C(R) + \alpha(\lambda - 1)^2 + \beta_t \frac{\|R - I\|_F^2}{C}$.
 - 12: Backpropagate and update A , λ .
 - 13: **End Loop**
 - 14: Final rotation (inline): $R^* = \exp(A - A^\top)$.
 - 15: Apply alignment (inline): $\tilde{L}_{i,a,e} = R^{*\top} L'_{i,a,e} R^*$.
 - 16: Exponential map (inline): $\tilde{C}_{i,a,e} = \exp(\tilde{L}_{i,a,e})$.
 - 17: **Output:** Pre-aligned covariances $\tilde{C}_{i,a,e}$.
-



Riemannian Fisher U-Net (RiFU) Pre-Aligner

Problem. Even after RA-whitening, covariances $C_{i,a,e} \in \mathbb{S}_{++}^C$ exhibit:

- ▶ subject-dependent orientation distortions,
- ▶ inconsistent action directions in tangent space,
- ▶ inflated subject-specific dispersion,
- ▶ degraded LOSO zero-shot performance.

RiFU is a *geometry-aware U-Net* that operates directly on SPD matrices using congruence transforms:

$$\Phi(C) = W^\top C W,$$

with skip connections merged via the log-Euclidean mean.

RiFU nonlinear encoder–decoder learns:

- ▶ tighter action clusters,
- ▶ suppressed subject variability,
- ▶ preserved SPD geometry,

yielding highly discriminative covariances for zero-shot decoding.



Riemannian Fisher U-Net (RiFU) Pre-Aligner

SPD Layers.

- Each encoder/decoder layer applies a congruence map:

$$\Phi(C) = W^\top C W, \quad W \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}.$$

Skip-Connection Merge (Log-Euclidean).

$$\text{merge}(C_1, C_2) = \exp\left(\frac{1}{2}(\log C_1 + \log C_2)\right)$$

U-Net Composition.

- Bottleneck

$$B = E(C') = \Phi^{(L)} \circ \dots \circ \Phi^{(1)}(C'),$$

- Transformed covariance

$$C^{\text{out}} = D(B) = \text{merge}\left(\Phi^{(d1)}(\text{merge}(\Phi^{(d2)}(B), C_1)), C'\right)$$

Tangent-Feature Representation.

$$S^{\text{out}} = \log C^{\text{out}}, \quad S^{\text{in}} = \log C', \quad z = \text{vec}(S^{\text{out}}), \quad z^{\text{in}} = \text{vec}(S^{\text{in}})$$



Riemannian Fisher U-Net (RiFU) Pre-Aligner

Fisher Statistics.

- ▶ class (action)-specific means (i), subject specific means (ii) and global means (iii)

$$(i) \mu_k = \frac{1}{n_k} \sum_{a=k} z_{i,a,e}, \quad (ii) \nu_s = \frac{1}{n_s} \sum_{i=s} z_{i,a,e}, \quad (iii) \mu = \frac{1}{n_a} \sum_k \mu_k, \quad \nu = \frac{1}{n_s} \sum_s \nu_s.$$

- ▶ Between-Action Variance (i) and Within-Action Variance (ii)

$$(i) B(A) = \frac{1}{N} \sum_k n_k \|\mu_k - \mu\|^2, \quad (ii) W(A) = \frac{1}{N} \sum \|z - \mu_k\|^2,$$

- ▶ Within-Subject Variance (i)

$$W(S) = \frac{1}{N} \sum \|z - \nu_s\|^2$$

Reconstruction Loss.

$$\text{Rec} = \frac{1}{N} \sum \|z - z^{\text{in}}\|_2^2$$

RiFU Loss.

$$\mathcal{L}_{\text{RiFU}} = \lambda_w W(A) - \lambda_{\text{bet}} B(A) - \lambda_{\text{sub}} W(S) + \lambda_{\text{rec}} \text{Rec}$$



Algorithm 7 Riemannian Fisher U-Net (RiFU) Pre-Aligner

- 1: **Input:** RA-whitened covariances $C'_{i,a,e}$; action labels $y_{i,a,e}^{\text{act}}$; subject labels $y_{i,a,e}^{\text{sub}}$.
 - 2: Initialize SPD U-Net parameters of encoder $E(\cdot)$ and decoder $D(\cdot)$ (all Φ -layers).
 - 3: **Optimization loop:** for each training minibatch
 - 4: *Forward pass:* $C_{i,a,e}^{\text{out}} = D(E(C'_{i,a,e}))$.
 - 5: Compute log-maps (inline): $S_{i,a,e}^{\text{out}} = \log C_{i,a,e}^{\text{out}}, \quad S_{i,a,e}^{\text{in}} = \log C'_{i,a,e}$.
 - 6: Vectorize (inline): $z_{i,a,e} = \text{vec}(S_{i,a,e}^{\text{out}}), \quad z_{i,a,e}^{\text{in}} = \text{vec}(S_{i,a,e}^{\text{in}})$.
 - 7: Compute Fisher stats (inline): $W(A), B(A), W(S), \text{Rec.}$
 - 8: Compute loss (inline): $\mathcal{L}_{\text{RiFU}}$.
 - 9: *Backpropagate:* update all W matrices in E, D via gradient descent.
 - 10: **End loop**
 - 11: Save E^*, D^* and apply final mapping (inline): $\tilde{C}_{i,a,e} = E^*(D^*(C'_{i,a,e}))$.
 - 12: **Output:** Pre-aligned covariances $\tilde{C}_{i,a,e}$.
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Classifier Module



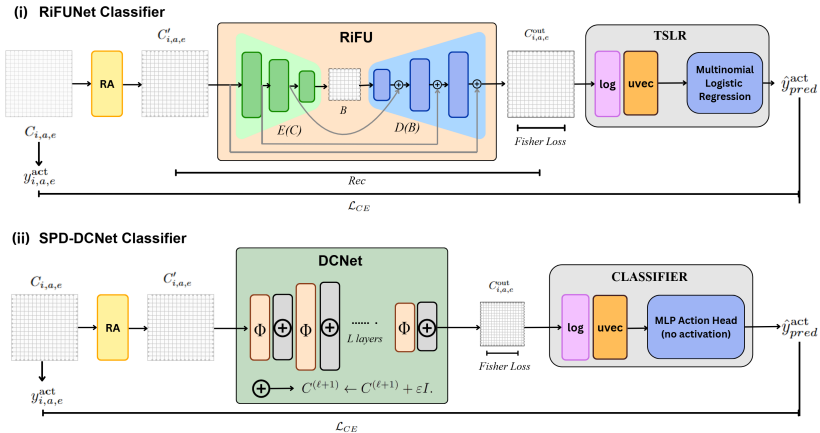


Figure 7: i) **RiFUNet** classifier with a Riemannian U-Net feature extractor followed by a TSLR action head. (ii) **SPD-DCNet** classifier composed of deep congruence layers with a linear (no-activation) action head. Together, these architectures illustrate Deep Congruence Networks for cross-subject action decoding.



RiFUNet Classifier (Riemannian Fisher U-Net)

Problem. Even after RA, motor-imagery covariances $C_{i,a,e} \in \mathbb{S}_{++}^C$ retain strong subject-induced orientation distortions that degrade supervised cross-subject decoding. RiFUNet extends the unsupervised RiFU Pre-Aligner into a *supervised* classifier by combining:

- ▶ a Riemannian U-Net feature extractor (congruence transforms),
- ▶ a custom TSLR tangent-space action head,
- ▶ cross-entropy loss for discriminative action learning,
- ▶ Fisher constraints to suppress subject-specific variance,
- ▶ optional reconstruction for geometric fidelity.

RiFUNet learns:

- ▶ discriminative action logits in tangent space,
- ▶ subject-invariant geometric features,
- ▶ SPD-preserving multiscale representations,

providing strong LOSO zero-shot generalization.



RiFUNet Classifier (Riemannian Fisher U-Net)

Congruence Transform Layers.

- Each U-Net layer applies:

$$C^{(\ell+1)} = W_\ell^\top C^{(\ell)} W_\ell, \quad C^{(\ell)} \in \mathbb{S}_{++}^{d_\ell}.$$

Skip-Connection Merge.

- Decoder merges use log-Euclidean averaging:

$$\text{merge}(C_1, C_2) = \exp\left(\frac{1}{2}(\log C_1 + \log C_2)\right),$$

preserving SPD geometry.



RiFUNet Classifier (Riemannian Fisher U-Net)

Weight Initialization.

- ▶ Let the euclidean global mean be $M = \frac{1}{N} \sum C'_{i,a,e}$ with eigendecomposition $M = U \Lambda U^\top$,
- ▶ Each W_ℓ is initialized as:

$$W_\ell = U[1:d_{\text{in}}, 1:d_{\text{out}}],$$

ensuring geometry-aware, stable activations.

Tangent-Space Features (TSLR).

- ▶ TSLR aligns each output covariance to the batch log-Euclidean mean:

$$(z_{i,a,e}, \hat{y}_{i,a,e}^{\text{act}}) = \text{TSLR}(C_{i,a,e}^{\text{out}}), \quad \mathcal{L}_{CE} = \frac{1}{N} \sum_{i,a,e} \text{CE}(\hat{y}_{i,a,e}^{\text{act}}, y_{i,a,e}^{\text{act}}).$$

producing upper-triangular tangent vector $z_{i,a,e}$ and action logits $\hat{y}^{\text{act}}$ enabling CE gradients to flow through TSLR and into all congruence layers.



RiFUNet Classifier (Riemannian Fisher U-Net)

Fisher Statistics.

- For actions discriminability and subject alignment:

$$W(A) = \frac{1}{N} \sum \|z - \mu_a\|_2^2, \quad B(A) = \frac{1}{N} \sum_a n_a \|\mu_a - \mu\|_2^2.$$

$$W(S) = \frac{1}{N} \sum \|z - \nu_s\|_2^2, \quad B(S) = \frac{1}{N} \sum_s n_s \|\nu_s - \nu\|_2^2.$$

Reconstruction Loss.

$$\mathcal{L}_{Rec} = \frac{1}{N} \sum \|C_{i,a,e}^{\text{out}} - C'_{i,a,e}\|_F^2.$$

Total Loss.

$$\mathcal{L}_{RC} = \lambda_{CE} \mathcal{L}_{CE} + \lambda_{act} (\lambda_w W(A) - \lambda_b B(A)) + \lambda_{sub} (\lambda_b B(S) - \lambda_w W(S)) + \lambda_{rec} \mathcal{L}_{Rec}.$$



Algorithm 8 RiFUNet Classifier

- 1: **Input:** RA-whitened covariances $C'_{i,a,e}$; action labels $y_{i,a,e}^{\text{act}}$; subject labels $y_{i,a,e}^{\text{sub}}$.
 - 2: Set hyperparameters λ_{CE} , λ_{act} , λ_w , λ_b , λ_{sub} , λ_{rec} .
 - 3: Initialize SPD linear layers W_ℓ in the RiFU U-Net.
 - 4: **Optimization loop:**
 - 5: *Forward pass:* $C_{i,a,e}^{\text{out}} = \text{RiFU}(C'_{i,a,e})$.
 - 6: Compute tangent features and logits (inline): $(z_{i,a,e}, \hat{y}_{i,a,e}^{\text{act}}) = \text{TSLR}(C_{i,a,e}^{\text{out}})$.
 - 7: Compute Fisher stats: $W(A)$, $B(A)$, $W(S)$, $B(S)$.
 - 8: Compute total loss: $\mathcal{L}_{RC}$.
 - 9: *Backpropagate:* update U-Net + TSLR weights.
 - 10: **End loop**
 - 11: Save trained E^* , D^* , forming RiFU * and TSLR * .
 - 12: Final forward pass: $\tilde{C}_{i,a,e}^{\text{out}} = \text{RiFU}^*(C'_{i,a,e})$.
 - 13: Tangent features and logits (inline): $(\tilde{z}_{i,a,e}, \hat{y}_{\text{pred}}^{\text{act}}) = \text{TSLR}^*(\tilde{C}_{i,a,e}^{\text{out}})$.
 - 14: **Output:** predicted labels $\hat{y}_{\text{pred}}^{\text{act}}$ and features $\tilde{z}_{i,a,e}$.
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SPD-DCNet Classifier (Deep Congruence Network)

Problem. Even after RA, EEG covariances $C_{i,a,e} \in \mathbb{S}_{++}^C$ retain subject-specific geometric distortions. RiFUNet addresses this with an encoder–decoder U-Net, but a lighter model may suffice.

SPD-DCNet is a *pure deep congruence network*: a feed-forward stack of SPD-preserving maps

$$C^{(\ell+1)} = W_\ell^\top C^{(\ell)} W_\ell,$$

with small SPD stabilization $C^{(\ell+1)} \leftarrow C^{(\ell+1)} + \varepsilon I$. It removes:

- ▶ decoder symmetry,
- ▶ skip connections,
- ▶ reconstruction constraints.

Despite this minimalism, SPD-DCNet achieves strong LOSO performance:

- ▶ deep congruence stacks alone produce discriminative SPD embeddings,
- ▶ Fisher penalties suppress subject structure,
- ▶ tangent-space linear head yields clean decision geometry.



SPD-DCNet Classifier (Deep Congruence Network)

Congruence Stack (Feed-Forward).

- ▶ Each layer applies:

$$C^{(\ell+1)} = W_\ell^\top C^{(\ell)} W_\ell, \quad C^{(\ell)} \in \mathbb{S}_{++}^{d_\ell}.$$

SPD Stabilization.

- ▶ To ensure numerical stability:

$$C^{(\ell+1)} \leftarrow C^{(\ell+1)} + \varepsilon I.$$

Width Patterns.

- ▶ Compact patterns such as:

$$d \rightarrow 2d \rightarrow 2d \rightarrow d$$

or extended variants $d-2d-3d-2d-d$ work well for LOSO decoding.



SPD-DCNet Classifier (Deep Congruence Network)

Weight Initialization.

- ▶ Let the euclidean global mean be $M = \frac{1}{N} \sum C'_{i,a,e}$ with eigendecomposition $M = U \Lambda U^\top$,
- ▶ Each W_ℓ is initialized as:

$$W_\ell = U[1:d_{\text{in}}, 1:d_{\text{out}}],$$

ensuring geometry-aware, stable activations.

Tangent-Space Vectorization.

- ▶ Final SPD matrix is mapped to log-domain and vectorized:

$$S_i = \log C_{i,a,e}^{\text{out}}, \quad z_{i,a,e} = \text{vec}_\Delta(S_i).$$

Linear Action Head (No Activations).

- ▶ A multilayer linear classifier processes tangent features:

$$\text{logits} = W_L W_{L-1} \cdots W_1 z_{i,a,e} + b, \quad \hat{y}_{\text{pred}}^{\text{act}} = \arg \max(\text{logits})$$

preserving the geometry established by the congruence stack.



SPD-DCNet (Deep Congruence Network)

Cross-Entropy Loss.

$$\mathcal{L}_{CE} = \frac{1}{N} \sum \text{CE}(\hat{y}_{pred}^{\text{act}}, y^{\text{act}}).$$

Fisher Statistics

- For actions discriminability and subject alignment:

$$W(A) = \frac{1}{N} \sum \|z - \mu_a\|^2, \quad B(A) = \frac{1}{N} \sum_a n_a \|\mu_a - \mu\|^2.$$

$$W(S) = \frac{1}{N} \sum \|z - \nu_s\|^2, \quad B(S) = \frac{1}{N} \sum_s n_s \|\nu_s - \nu\|^2.$$

Total Loss (No Reconstruction Term).

$$\mathcal{L}_{DC} = \lambda_{CE} \mathcal{L}_{CE} + \lambda_{act} (\lambda_w W(A) - \lambda_b B(A)) + \lambda_{sub} (\lambda_b B(S) - \lambda_w W(S)).$$

Note. Unlike RiFUNet, SPD-DCNet *omits reconstruction* to focus entirely on discriminative SPD geometry.



Algorithm 9 SPD-DCNet Classifier

- 1: **Input:** RA-whitened covariances $C'_{i,a,e}$; action labels $y_{i,a,e}^{\text{act}}$; subject labels $y_{i,a,e}^{\text{sub}}$.
 - 2: Initialize $\{W_1, \dots, W_L\}$ stacks and ActionHead.
 - 3: **for** each training iteration **do**
 - 4: *Forward pass:* $C^{(1)} = C'_{i,a,e}$.
 - 5: **for** $\ell = 1 \dots L$ **do**
 - 6: $C^{(\ell+1)} = W_\ell^\top C^{(\ell)} W_\ell$.
 - 7: $C^{(\ell+1)} \leftarrow C^{(\ell+1)} + \varepsilon I$.
 - 8: **end for**
 - 9: $C_{i,a,e}^{\text{out}} = C^{(L+1)}$.
 - 10: Compute log-domain features: $S_i = \log C_{i,a,e}^{\text{out}}$, $z_i = \text{vec}_\Delta(S_i)$.
 - 11: Predict $\hat{y}_{\text{pred}}^{\text{act}} = \text{ActionHead}(z_{i,a,e})$.
 - 12: Compute Fisher stats: $W(A)$, $B(A)$, $W(S)$, $B(S)$.
 - 13: Compute total loss $\mathcal{L}_{DC}$.
 - 14: *Backpropagate:* update $\{W_\ell\}$ and ActionHead.
 - 15: **end for**
 - 16: Save $\{W_\ell\}^*$ and ActionHead*.
 - 17: *Final forward pass:* $C^{(1)} = C'_{i,a,e}$; repeat congruence stack to obtain $\tilde{C}_{i,a,e}^{\text{out}}$.
 - 18: Compute $\tilde{S}_{i,a,e} = \log \tilde{C}_{i,a,e}^{\text{out}}$, $\tilde{z}_{i,a,e} = \text{vec}_\Delta(\tilde{S}_{i,a,e})$
 - 19: Predict $\hat{y}_{\text{pred}}^{\text{act}} = \text{ActionHead}^*(z_{i,a,e})$.
 - 20: **Output:** predicted labels $\hat{y}_{\text{pred}}^{\text{act}}$ and features $z_{i,a,e}$.
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Preprocessing Module

As shown in Table 1, the RiFu PreAligner provides the most consistent gains across classifiers, offering the largest improvement for TSLR (+1.9%) and also enhancing TSA-LDA. Its log-domain filtering and skip-connected smoothing reduce subject-specific covariance distortions more effectively than RA.

Table 1: Benchmarking of preprocessing modules under LOSO cross-subject evaluation on BCI-IV 2a. Mean classification accuracy (%) is reported.

Classifier	RA	DCR	RiFu
TSLR	52.89 ± 14.48	53.01 ± 16.71	54.78 ± 15.43
TSA-LDA	53.51 ± 15.29	53.97 ± 16.02	54.24 ± 15.81
MDM	52.43 ± 15.66	50.46 ± 14.82	52.89 ± 15.41

DCR provides only modest improvements for TSA-LDA and TSLR but degrades MDM, reflecting its sensitivity to rotation-only transforms and the stricter geometric preservation required by metric-based classifiers. Overall, RiFu consistently outperforms both RA and DCR, making it the most stable and robust preprocessing strategy for cross-subject EEG decoding.



Classifier Module

As shown in Table 2, classical baselines such as MDM, TSLR, and TSA-LDA provide reasonable LOSO performance, with TSA-LDA being the strongest. CSP-based models perform considerably worse due to channel noise and non-stationarity, although z-score normalization provides small improvements.

Table 2: Subject-wise LOSO accuracy (%) across baselines and deep classifiers. Best per-subject scores are underlined, overall best is in **bold**.

Subject	MDM	TSLR	TSA-LDA	CSP-LDA	CSP-LDA-Z	RiFuNet	SPD-DCNet
A01T	61.81	61.46	67.71	53.47	58.68	67.36	<u>68.40</u>
A02T	26.39	29.51	28.82	24.31	25.35	27.08	<u>29.86</u>
A03T	72.92	64.93	73.96	59.03	65.97	80.21	<u>82.64</u>
A04T	44.79	44.44	<u>47.92</u>	35.76	39.58	44.79	<u>43.06</u>
A05T	42.71	38.54	43.40	31.25	27.78	<u>45.49</u>	<u>44.79</u>
A06T	32.29	<u>42.36</u>	37.50	25.00	24.31	41.67	<u>40.28</u>
A07T	<u>59.38</u>	45.49	47.22	28.82	52.78	50.00	<u>54.51</u>
A08T	71.18	68.06	72.57	70.83	64.58	<u>76.74</u>	<u>76.39</u>
A09T	60.42	60.42	62.50	50.69	51.39	62.85	<u>65.62</u>
Mean $\pm$ Std	52.43 $\pm$ 15.66	50.58 $\pm$ 12.68	53.51 $\pm$ 15.29	42.13 $\pm$ 15.84	45.60 $\pm$ 15.81	55.13 $\pm$ 16.66	<u>56.17 $\pm$ 16.99</u>

Both proposed deep geometric models outperform classical baselines. **SPD-DCNet** achieves the highest mean accuracy (56.17%), demonstrating the effectiveness of hierarchical congruence transforms for discriminative SPD representations. **RiFuNet** performs almost as strongly (55.13%), with peak performance on subjects A03T, A08T, and A09T. Overall, deep congruence networks provide a +3–4% improvement over the best classical baselines, confirming superior cross-subject generalization.



Thank You

